The Journal of Chemical Physics

RESEARCH ARTICLE | JANUARY 02 2020

Quantum control of radical pair reactions by local optimization theory

Special Collection: Spin Chemistry

Kenta Masuzawa ; Masaya Sato; Michihiko Sugawara; Kiminori Maeda

Check for updates

J. Chem. Phys. 152, 014301 (2020)
https://doi.org/10.1063/1.5131557

View Online Export Citation

Articles You May Be Interested In

Analytical model for spin dynamics in radical pairs under the spin-locking condition

J. Chem. Phys. (June 2024)

Forbidden transitions and the effective masses of electrons and holes in In 1−x Ga x As/InP quantum wells with compressive strain

J. Appl. Phys. (January 2003)

Quantum dynamics driven by continuous laser fields under measurements: Towards measurement-assisted quantum dynamics control

J. Chem. Phys. (November 2005)

14 May 2026 17:45:38

AIP Advances
Why Publish With Us?
21DAYS average time to 1st decision
OVER 4 MILLION views in the last year
INCLUSIVE scope
Learn More
AIP Publishing

AIP Publishing

The Journal of Chemical Physics ARTICLE scitation.org/journal/jcp

Quantum control of radical pair reactions by local optimization theory

Cite as: J. Chem. Phys. 152, 014301 (2020); doi: 10.1063/1.5131557
Submitted: 14 October 2019 • Accepted: 12 December 2019 •
Published Online: 2 January 2020

View Online Export Citation CrossMark

Kenta Masuzawa,[1] Masaya Sato,[1,a)] Michihiko Sugawara,[2] and Kiminori Maeda[1,b)]

AFFILIATIONS

[1] Department of Chemistry, Graduate School of Science and Engineering, Saitama University, 255 Shimo-okubo, Sakura Ward, 338-8570 Saitama, Japan
[2] Keio Quantum Computing Center, Keio University, Kanagawa 223-8522, Japan

Note: This paper is part of the JCP Special Topic on Spin Chemistry.
[a)] **Present address:** Graduate School of Arts and Sciences, The University of Tokyo, Komaba, 153-8902 Tokyo, Japan.
[b)] **Author to whom correspondence should be addressed:** kiminorimaeda@mail.saitama-u.ac.jp

ABSTRACT

Recently, AWG (arbitrary waveform generator) based pulse electron paramagnetic resonance and nuclear magnetic resonance have been developed in a high field regime for the improvement of sensitivity and selectivity and quantum information processing. Here, we propose the application of AWG based reaction control of radical pairs in a rather low magnetic field regime. We calculated the locally optimized radio frequency (RF) field with the control theory by Sugawara [J. Chem. Phys. **118**(15), 6784–6800 (2003)]. The calculation results manifest the applicability of AWG-RF fields to reaction control (reaction yield detected magnetic resonance), stimulated nuclear polarization, magnetic isotope selection, and coherent control of the spin dynamics.

Published under license by AIP Publishing. https://doi.org/10.1063/1.5131557

14 May 2026 17:45:38

I. INTRODUCTION

In spin chemistry, radical pair (RP) dynamics have been mostly studied by static and oscillating magnetic field effects (MFEs) and electron and nuclear spin polarizations.[1] The recent development of the study on organic semiconductor and solar cells provides the extension of the concept of radical pair theory to the spin pairs in solid state material science.[2–5] In life science and biology, radical pair concepts have also gained attraction from the hypothesis that the RP generated in the retina of birds acts as a magnetic compass-based sensor.[6] One of the most important pieces of circumstantial evidence for this hypothesis is the radio wave oscillating magnetic field effect on bird orientation.[7–9] Such phenomena can be explained by a concept called reaction yield detected magnetic resonance (RYDMR).[10] In spin chemistry, RYDMR has been observed by various techniques, such as fluorescence,[11–13] absorption spectroscopy,[14,15] electric conductivity,[16–19] spin trapping,[20–22] and liquid chromatography.[23]

RYDMR is based on the modification of the reaction yield by the magnetic resonance and the spin selective chemical reaction as shown in Fig. 1. If RPs are generated from a singlet excited state, the RPs are initially in the singlet manifold. Under a high magnetic field condition, singlet RPs are effectively mixed with the T_0 spin state via the hyperfine interaction. If the oscillating field is resonant to one of the electron spins of the RP, the transition from S-T_0 mixed states to T_+ and T_- states, which are isolated from the singlet state by Zeeman splitting, rescues the RP population to the T_+ and T_- states from the charge recombination reaction. Therefore, the radio frequency (RF) field induced transition provides a longer RP lifetime.[24] In the case of triplet-born RPs, the direction of the transition is opposite and the RF accelerates the charge recombination process.

The RF resonance provides not only population transfer but also coherent quantum mechanical spin motions, such as spin locking or Rabi oscillation. Therefore, the dynamics of the spin motion is not simply characterized by the transition kinetics based on the Fermi Golden rule and is more complicated in multinuclear systems.

Here, we present arbitrary waveform generator (AWG) based RYDMR. This concept is similar to previously developed laser-based reaction control techniques.[25–27] If we make a comparison between

Published under license by AIP Publishing

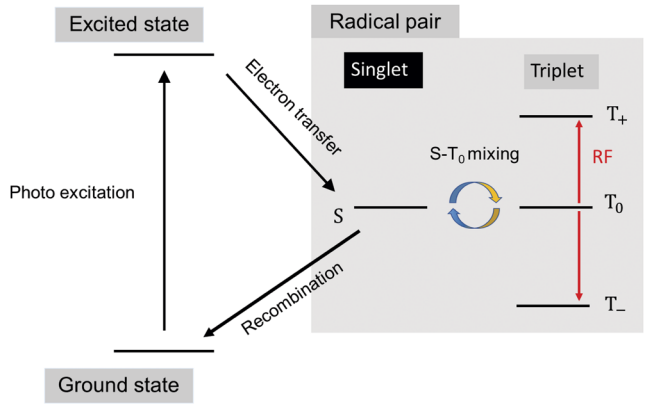


FIG. 1. Reaction scheme and schematic level diagram of model radical pairs. Pumping of the population from T_0 to T_+, T_- states avoids fast geminate recombination reaction. Therefore, the RP lifetime is expected to become longer by RF irradiation.

the radio wave- and laser-based reaction control techniques, we can comment on the following:

1. The photon energy of RF or microwave (MW) field is extremely small compared to the thermal energy, i.e., the typical driving force of chemical reactions. Therefore, the RP-based magnetoreception and RYDMR effect are pure quantum mechanical phenomena.
2. The production of an AWG RF pulse is technically much easier and more cost effective than laser shaping and phase control.
3. In photochemistry, RPs are produced from singlet or triplet precursor excited states. Therefore, the initial conditions of RP electron spins are highly polarized and entangled.
4. An electron spin system is not directly coupled with the thermal bath. Therefore, the spin relaxation and dephasing time is much longer than that of typical vibronic or electronic states. Therefore, one can control the chemical reaction by manipulating the quantum systems on a relatively long time scale.

The above motivated us to try AWG based reaction control.

The theory for designing electromagnetic fields has been developed independently in the fields of both Nuclear Magnetic Resonance (NMR) and laser pulse shaping. In the NMR field, the gradient ascent of parameter estimation (GRAPE) has been widely used.[28] These calculations are based on the so-called "global optimization" method. However, these global searches of the optimized parameter sets need an iterative procedure for the all-time region, and sometimes, it needs the trial with various initial conditions in the optimizing procedure. Concerning it, at present, we applied the rather simple technique called "local optimization." This technique allows us to calculate the time dependent RF signal without an iterative procedure satisfying the monotonous increase of the locally defined general performance index, $y(t)$. Besides the large merit that the calculation is more cost effective, one can calculate smooth RF shapes with smaller time resolution avoiding the discrete jump of the RF wave forms. For general RP systems, one should take into account all nuclear spin configurations, which produces a very large density matrix. However, this problem can be overcome simply by solving a simultaneous equation under the high field approximation.

II. THEORY

The optimization is based on the methods developed by Sugawara.[25] This theory is built for the quantum control of the chemical system by shaped infrared laser light. However, the theory and the concept should be applicable for general quantum systems. Here, we reformulate the local optimization theory for controlling RPs. The spin Hamiltonian of RPs is written by

$$\hat{H}_0 = \hat{H}_A + \hat{H}_B + 2J\left(\frac{1}{2} + 2\hat{S}_A \bullet \hat{S}_B\right), \tag{1}$$

where the first and second terms are corresponding to radical A and radical B, respectively, and the third term is the exchange interaction of the radicals in the RP. The Hamiltonians for radicals A, B are written by the electron spin Zeeman and hyperfine interactions as shown in

$$\begin{aligned} \hat{H}_A &= g_A\mu_B B_0 \hat{S}_Z^A + \sum_{i\in A} a_i \hat{S}^A \bullet \hat{I}_i, \\ \hat{H}_B &= g_B\mu_B B_0 \hat{S}_Z^B + \sum_{j\in B} a_j \hat{S}^B \bullet \hat{I}_j. \end{aligned} \tag{2}$$

Under the high field approximation, electron spins $\hat{S}^{A,B}$ are quantized to the direction of the external magnetic field, Z, and nuclear spin operators $\hat{I}_{i,j}$ can be replaced by the nuclear spin quantum numbers m_i, n_j for the nuclei coupled with the electron spins. Now, the Hamiltonians for radicals A, B are given by

$$\begin{aligned} \hat{H}_A^m &= \left(g_A\mu_B B_0 + \sum_{i\in A} a_i m_i\right)\hat{S}_Z^A = \omega_A^m \hat{S}_Z^A, \\ \hat{H}_B^n &= \left(g_B\mu_B B_0 + \sum_{j\in B} a_j n_j\right)\hat{S}_Z^B = \omega_B^n \hat{S}_Z^B, \end{aligned} \tag{3}$$

where $m = \{m_i\}$, $n = \{n_j\}$ represents the nuclear configuration of radical A and B, respectively. Now, we put a new index $k = \{m, n\}$. Considering the radical pair system with Hamiltonian $\hat{H}_0^k$,

$$\hat{H}_0^k = \hat{H}_A^m + \hat{H}_B^n + J\left(\frac{1}{2} + 2\hat{S}_A \bullet \hat{S}_B\right). \tag{4}$$

The eigenstates of the Hamiltonian can be calculated by the equation

$$\hat{H}_0^k |l,k\rangle = E_0^k |l,k\rangle, \tag{5}$$

and the solutions are[29]

$$\begin{aligned} &|1,k\rangle = |T_{+1},k\rangle, \quad E_1^k = +\left(\omega_A^m + \omega_B^n\right)/2 + J, \\ &|2,k\rangle = \cos\theta|S,k\rangle + \sin\theta|T_0,k\rangle, \quad E_2^k = +\sqrt{q_k^2 + J^2}, \\ &|3,k\rangle = \sin\theta|S,k\rangle - \cos\theta|T_0,k\rangle, \quad E_3^k = -\sqrt{q_k^2 + J^2}, \\ &|4,k\rangle = |T_{-1},k\rangle \quad E_4^k = -\left(\omega_A^m + \omega_B^n\right)/2 + J, \end{aligned} \tag{6}$$

14 May 2026 17:45:38

Published under license by AIP Publishing

where

$$q_k = \left(\omega_A^m - \omega_B^n\right)/2, \quad \tan 2\theta = \frac{q_k}{J}. \tag{7}$$

The eigenvectors $|l, k\rangle$can be classified into two groups. One group includes $|T_{+1}, k\rangle$and $|T_{-1},k\rangle$. The other group includes $|2, k\rangle$and $|3, k\rangle$, which contains a singlet character in the wavefunctions.

Under the high-field approximation, there is no mixing nor coherence between the different nuclear configurations, k. Therefore, the total density vector $|\rho(t)\rangle_L$ in the Liouville space can be divided into the vector for the individual nuclear configuration, $|\rho_k(t)\rangle_L$.

The performance index, $y(t)$, is defined by

$$y(t) = \sum_k q_k \langle P_k|\rho_k(t)\rangle_L, \tag{8}$$

P_k is the projection operator for the target states and the weighting factor q_k provides the priority for the target states P_k. If singlet character of the RP effectively reacts to the ground state, the design of the RF field targeting the eigenstates having large singlet character,

$$\hat{P}_k = |2,k\rangle\langle 2,k| + |3,k\rangle\langle 3,k|, \tag{9}$$

accelerates the chemical reaction from singlet RPs. In contrast, if we use

$$\hat{P}_k = |1,k\rangle\langle 1,k| + |4,k\rangle\langle 4,k|, \tag{10}$$

RPs can be stored in the nonreactive state. Therefore, the lifetime of RPs is manipulated by the optimized RF field. For optimization of the external oscillating field $u(t)$, time evolution of the system is described by the Liouville von Neuman equation

$$\frac{d|\rho_k(t)\rangle_L}{dt} = -i\hat{\hat{L}}_0^k|\rho_k(t)\rangle_L - i\hat{\hat{L}}_V u(t)|\rho_k(t)\rangle_L \tag{11}$$

where, $\hat{\hat{L}}_0$, $\hat{\hat{L}}_V$, are super operators presenting commutators in the Hilbert space, as shown,

$$\hat{\hat{L}}_0 = \hat{H}_0 - \tilde{H}_0, \quad \hat{\hat{L}}_v = \frac{g_{A,B}\mu_B}{\hbar}\left(\hat{S}_X - \tilde{S}_X\right), \tag{12}$$

where hat and tilde indicate the operation from left and right, respectively, in the Hilbert space.

Time derivative of $y(t)$ is

$$\begin{aligned}\frac{dy}{dt} &= \frac{d\sum_k q_k\langle P_k|\rho_k(t)\rangle_L}{dt} \\ &= \sum_k q_k\langle P_k|\left\{-i\hat{\hat{L}}_0^k|\rho_k(t)\rangle_L - iu(t)\hat{\hat{L}}_v|\rho_k(t)\rangle_L\right\} \\ &= -i\sum_k q_k\left\{\langle P_k|\hat{\hat{L}}_0^k|\rho_k(t)\rangle_L + u(t)\langle P_k|\hat{\hat{L}}_v|\rho_k(t)\rangle_L\right\}.\end{aligned} \tag{13}$$

The first term in the summation is eliminated by using the relationship $\left[\hat{H}_0, \hat{P}_k\right] = 0$.

Now, we introduce an amplitude parameter A and when the external oscillating magnetic field $u(t)$ is given by

$$u(t) = iA\sum_k q_k\langle P_k|\hat{\hat{L}}_V|\rho_k(t)\rangle_L \quad (A > 0), \tag{14}$$

a positive time derivative of $y(t)$ is guaranteed

$$\frac{dy(t)}{dt} = A\left[\sum_k q_k\langle P_k|\hat{\hat{L}}_V|\rho_k(t)\rangle_L\right]^2 \geq 0. \tag{15}$$

Now, we put (14) into (11) and obtain the nonlinear simultaneous equation

$$\frac{d|\rho_k(t)\rangle_L}{dt} = -i\hat{\hat{L}}_0^k|\rho_k(t)\rangle_L + \hat{\hat{L}}_V\left\{A\sum_j q_j\langle P_j|\hat{\hat{L}}_V|\rho_j(t)\rangle_L\right\}|\rho_k(t)\rangle_L. \tag{16}$$

This simultaneous equation can be solved numerically using the Runge-Kutta method. Practically, at a discrete time t_n, we solved Eq. (16) of $|\rho_k(t_n)\rangle$and obtained the next density vector $|\rho_k(t_{n+1})\rangle$for all k simultaneously. Using $|\rho_k(t_{n+1})\rangle$, the desired RF field $u(t_{n+1})$ is calculated by Eq. (14). The time step $\Delta t = t_{n+1} - t_n$ is automatically determined by the Runge-Kutte numerical package of the software.

Until here, the target spin states are limited to the eigenstates of the original spin Hamiltonian because the commutating relationship $\left[\hat{H}_0, \hat{P}_k\right] = 0$ is a necessary condition for the derivation of Eq. (14). However, the coherent control targeting the noneigenstates is also possible as follows: In principle, noneigenstates of the Hamiltonian automatically evolve in time. Therefore, we need to set the targeting states on a specific time t_f, $P_k(t_f)$. Now, we use reverse time evolution

$$|P_k(t)\rangle_L = \exp\left\{i\hat{\hat{L}}_0^k\left(t_f - t\right)\right\}|P_k\left(t_f\right)\rangle_L. \tag{17}$$

The coherent control can be operated by replacement of ${}_L\langle P_j|$ by

$${}_L\langle P_j(t)| = {}_L\langle P_j\left(t_f\right)|\exp\left\{i\hat{\hat{L}}_0^k\left(t - t_f\right)\right\} \tag{18}$$

in Eq. (16).

Calculations were all executed with the computer language MATLAB 2016b (Mathworks) on a laptop computer (Microsoft, Surface Pro 3, Core i5 CPU Dual Core). The calculation time was about ten min for one optimization of RF field.

III. RESULTS AND DISCUSSION

A. Model system

We used a model system with 5 nuclear spins (I = 1/2). Three are coupled with the electron spin in radical A, with hyperfine coupling constants (HFCs), 0.2, 0.5, 1.0 mT, and two nuclei couple with radical B, having HFCs, 0.2, 0.3 mT. The Zeeman energy, in frequency units, $g_A\mu_B B_0/h$, $g_B\mu_B B_0/h$ is fixed to 100 MHz for both radicals. The stick diagrams of $\omega_A^m/2\pi$, $\omega_B^n/2\pi$, which indicate Electron Paramagnetic Resonance (EPR) transition frequencies for the nuclear configurations of radicals A and B, are calculated under the high field approximation as shown in Fig. 2. The exchange interaction J was fixed to −1.0 MHz.

In this frequency region, the high field approximation is not perfectly held. However, the high field approximation has large merit for typical multinuclear spin systems. In the system with p nuclear spins (I = 1/2), the full quantum mechanical calculation without the high field approximation needs a density vector

14 May 2026 17:45:38

Published under license by AIP Publishing

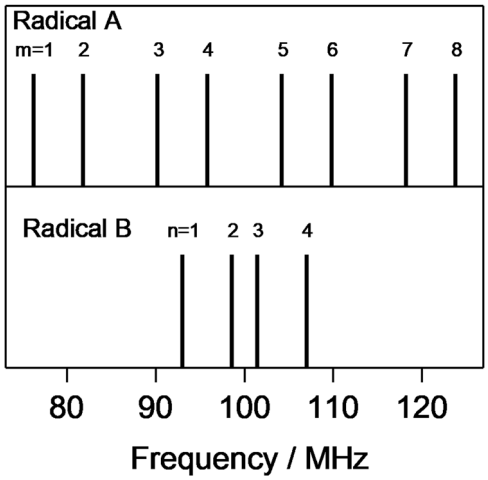


FIG. 2. Stick diagram of the resonance frequencies for the model RP system. Three nuclear spins couple with the electron spin in radical A with HFCs, 0.2, 0.5, 1.0 mT and two nuclei couple with radical B having HFCs, 0.2, 0.3 mT.

dimension $(2^2 \times 2^p)^2$ in the Liouville space. For example, it is 16 384 for the case of p = 5. In contrast, under the high field approximation, the dimension for one nuclear configuration can be reduced to 16 and 32 sets of nuclear configurations. Therefore, the dimension of the simultaneous Eq. (16) can be reduced to 512. The high field approximation has larger merit than the reduction in number of nuclear spins because (1) typical radical pairs contain many nuclear spins and the resonance frequencies of hyperfine structure are the most crucial factor for the optimization and the small mixing of the T_+ and T_- states with S-T_0 state by the violation of high field approximation is a secondary effect, and (2) the contribution of the forbidden transitions is small enough to be neglected.[30]

Some improvement of a rather crude high field approximation could be achieved with a calculation of the resonance frequencies by diagonalization of the exact spin Hamiltonian and the optimized RF field would be calculated using divided subensembles, ignoring the contribution of minor state mixing. However, this improvement has little consequence on the conclusion of the present manifest calculation. At present, the simplest high field approximation has been chosen for the calculation.

B. Population control of the RP system under high field approximation

At first, we attempted the population control for singlet born RPs in multinuclear systems. The optimized RF field is shown in Fig. 3(a) and the performance index $y(t)$ is monotonically increased by RF field irradiation and reached about 88% at around 2 μs. As shown in the inset of Fig. 3(a), the optimized RF field is quite smooth because we could calculate using a very small time step $\Delta t \sim 350$ ps [it is determined automatically by the ordinary differential equation (ODE) solver in MATLAB]. Figure 3(b) shows the time evolution of $\langle P_k \rangle$for the individual nuclear configuration $k = \{m, n\}$. In the nuclear configuration of radical A, $m = 4$ and 5, the rise in

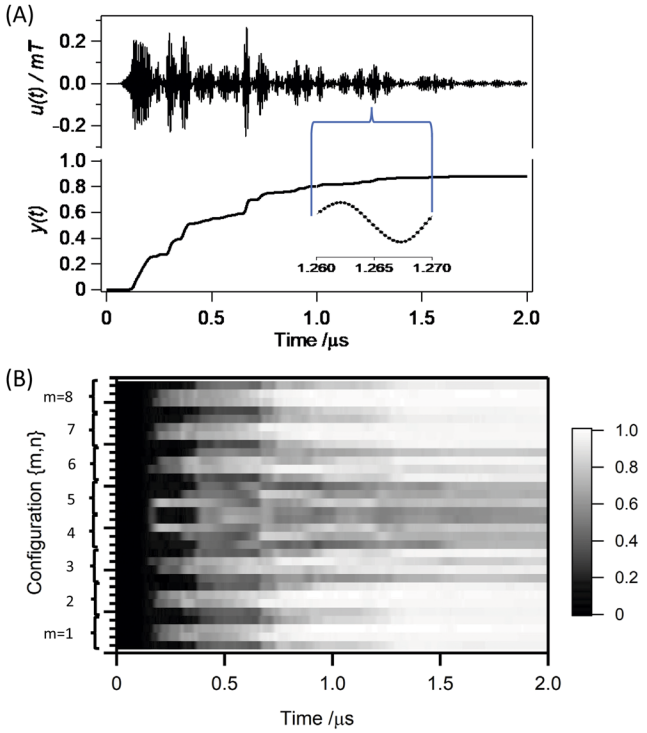


FIG. 3. (a) Calculation result of the optimized RF field $u(t)$ and the time evolution of the performance index $y(t)$. Zoom of $u(t)$ for the very narrow time window (10 ns) is inserted in figure. (b) Target probability of the individual nuclear configurations. Configurations {m, n}are in order of {1,1}, . . ., {1,n}, . . ., {2,1}, . . ., {m,n}.

probability of P_k states are slower than other configurations. This fact can be explained by slow S-T_0 spin mixing caused by small $q_k = (\omega_A^m - \omega_B^n)/2$, which is a rate determining step for transitions to T_+ and T_- spin states. The FFT of the optimized field shown in Fig. 4 gives us the information of the excitation feature of the hyperfine lines. The splitting of the FFT spectra around the hyperfine lines is given by the J coupling ($J = -0.1$ mT) and indicates the preciseness of the frequency optimization for the transitions.

C. Addition of the chemical reaction operator and the control of kinetics

In the part (b), the population control of the RP spin states was presented. Now, we study the effect of the optimized RF field on the reaction kinetics itself. For the purpose, we introduce the additional reaction superoperator, $\hat{\hat{K}}$,[31,32] given by

$$\hat{\hat{K}} = -\frac{k_S}{2}\left(\hat{P}_s + \tilde{P}_s\right) \tag{19}$$

and add the term $\hat{\hat{K}}|\rho_k(t)\rangle_L$ to Eqs. (11) and (16). In this model, the density matrix is no longer normalized,

$$\langle \rho | \rho \rangle_L \leq 1. \tag{20}$$

14 May 2026 17:45:38

Published under license by AIP Publishing

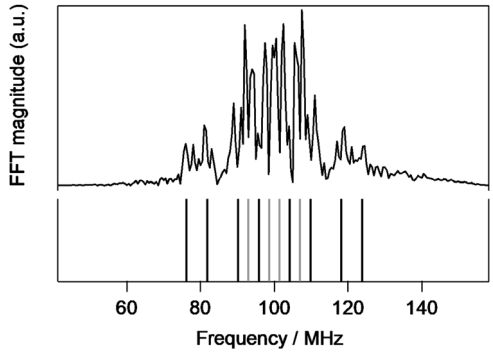


FIG. 4. FFT spectrum of the optimized RF field $u(t)$. The stick diagram of radical A (thick sticks) and B (thin sticks) found below for comparison with the spectrum.

However, the damped population (probability), $1 - \langle\rho|\rho\rangle_L$, can be regarded as the amount of product by spin selective chemical reaction, implicitly.[33,34] At the present calculation, we put $k_S = 2 \times 10^6\ s^{-1}$ and calculated $\langle\rho|\rho\rangle_L$ that represents the population fraction of radical pair, which is experimentally observable by various time resolved spectroscopic methods such as transient absorption (TA). Figure 5 shows that the optimized RF wave remarkably makes the RP lifetime longer. In contrast, single color RYDMR, in which the frequency of the RF is fixed at the center of the stick diagram (f = 100 MH), once accelerates the S-T mixing by the population transfer from S-T_0 mixed states to T_+ and T_- states. However, the strong RF field decelerates the spin mixing by spin locking.[14,15,35]

D. Nuclear spin state selection

Until now, we have used $q_k = 1$ for all nuclear configurations homogeneously. However, the selection and choice of q_k has various applications for the selection of subensembles such as nuclear spin configuration, anisotropic selection, and isotope selection. Here, we present one of the applications, nuclear spin selection. The nuclear selection by RF irradiation in RPs provides a nuclear polarization in NMR spectra of the products, which is called Stimulated Nuclear Polarization (SNP), and was developed by Sagdeev and Bagryanskaya *et al.*[36–39] For a test calculation, we tried a selection of nuclear spin states, n = 1, 3 in radical B nonselecting n = 2, 4, Fig. 2. Now, we set the weighting parameter array for radical B, $\{q_n^B\} = \{1, -1, 1, -1\}$, and for radical A, $\{q_m^A\} = \{1, 1, 1, 1, 1, 1, 1, 1\}$. The total parameter array for the radical pair nuclear configurations, k, is $\{q_k\} = \{q_m^A q_n^B\} = \{1, -1, 1, -1, 1, -1, \ldots\}$. It maximizes the nuclear polarization of the nucleus having the smallest hyperfine coupling constant a = 0.2 mT. The parameter $q_k = -1$ causes the penalty for the transition of undesired subensembles.

Figure 6(a) shows the desired RF field by the optimization and the time evolution of the total population of T_+ and T_- spin states (black line) and the performance index $y(t)$ (gray line). The smaller $y(t)$ compared to the population of T_+ and T_- spin states is caused by the penalty using undesired pumping. However, in total, the optimization is working very well and $y(t)$, which is regarded as SNP, is monotonically increased. The population of T_+ and T_- spin states are alternatively changed by the nuclear configuration, k, as shown in Fig. 6(b). The FFT spectrum is shown in Fig. 7. It is clearly seen

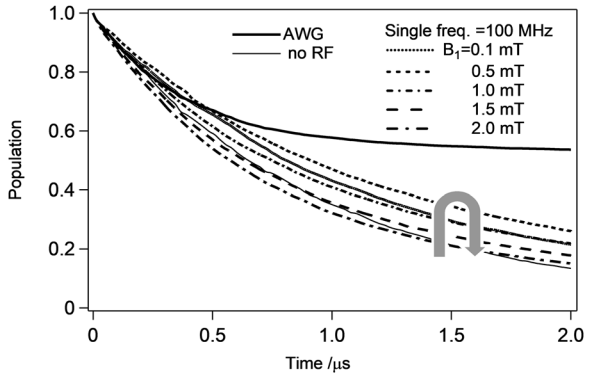


FIG. 5. Comparison of the AWG based RYDMR kinetics and the single colour RYDMR results. The gray arrow is the direction of RF power dependence (weak to strong RF power).

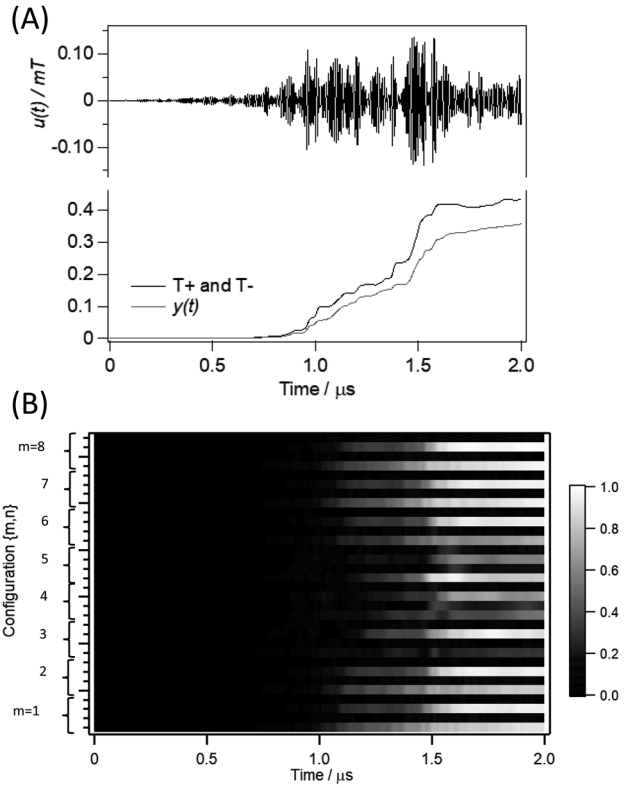


FIG. 6. Results of the nuclear spin configuration selective calculation. The condition is given in the text. (a) optimized RF field $u(t)$, the population of the target states, and the performance index $y(t)$. (b) The time evolution of the target probability of individual nuclear spin sub-ensembles. Configurations {m,n} are in order of {1,1}, . . .{1,n}, . . ., {2,1}, . . ., {m,n}.

14 May 2026 17:45:38

Published under license by AIP Publishing

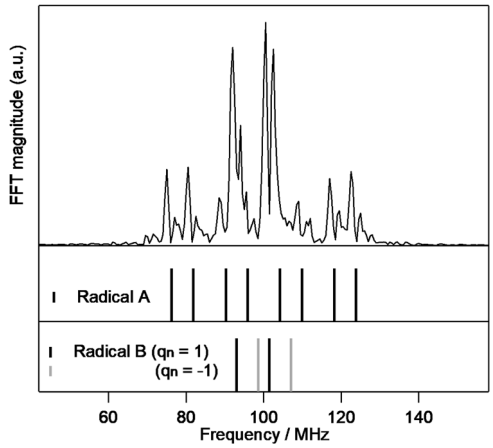


FIG. 7. FFT spectrum of the nuclear spin selective RF wave with the stick diagrams. Black and gray sticks of radical B corresponding to the frequencies with $q_n = 1$ and $q_n = -1$, respectively.

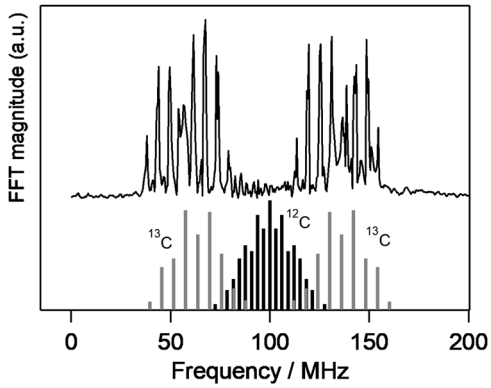


FIG. 8. FFT spectrum of the isotope selective RF wave with the stick diagrams. We put $q_k = 1$ for $k \in {}^{12}C$ and $q_k = -1$ for $k \in {}^{13}C$. In stick diagrams, black sticks and gray sticks represent the benzophenone ketyl radical containing ^{12}C and ^{13}C, respectively.

that the RF field tried to pump the transition for n = 1, 3 (black stick of radical B) avoiding the transition for n = 2, 4 (gray stick). It is surprising that there are weak transition components around 80 MHz and 120 MHz, which must be assigned as the transition of radical A. In principle, the transition of radical A had no selectivity of the nuclear states in radical B. Therefore, these transitions should not influence $y(t)$. Provably, these transitions reset the T_+ and T_- spin states of undesired nuclear configurations back to the S-T_0 mixed states and repump the population to the T_+ and T_- spin states using the resonance frequency of desired nuclear configuration of radical B. It is remarkable that such strategy can be automatically taken even with the local optimization method. We have also succeeded in the calculation using $\{q_n^B\} = \{1, 1, -1, -1\}$. The results are shown in the supplementary material.

E. Practical calculations for carbon isotope enrichments in triplet born radical pair kinetics

Not only for nuclear polarization but the optimization of the RF field can be used for the magnetic isotope effect in RP chemical reactions.[40–43] At present, we have used a more realistic but complicated model than used before, benzophenone in a micelle of Brij35 (Polyoxyethyleneglycol Dodecyl Ether). It is known that the photoinduced hydrogen abstraction reaction from triplet benzophenone produces a radical pair consisting of a benzophenone ketyl radical and a polyethylene glycol radical.[44] It is the case of a triplet born RP and the adduct of the RP is produced by geminate recombination from singlet RPs. In the present case, the RF transition accelerates the RP geminate recombination. The stick diagram of the RP using the hyperfine coupling constants (see the supplementary material) is shown in Fig. 8. The hyperfine coupling constant of ^{13}C at the center of the ketyl radical causes splitting of about 2.2 mT. In this calculation, we used $J = -0.01$ mT.

In the same manner as part (d), we put $q_k = 1$ for $k \in {}^{13}C$ and $q_k = -1$ for $k \in {}^{12}C$. As shown in Fig. 8, the RF field is resonant with, but avoiding, the transition of ^{12}C components. The RP decay kinetics are shown in Fig. 9. The RF field makes the transition from T_+ and T_- states to S-T_0 mixed states selectively for the ^{13}C containing RPs. Therefore, the decay of ^{13}C containing RPs is largely accelerated. It results in the condensation of ^{13}C on the adducts (recombination product) of RPs.

F. Coherent manipulation of RP spin dynamics

The coherent manipulation of RPs containing many nuclear spins is also possible even if it is a rather hard challenge. We have used the simplified system as shown in the stick diagram of Fig. 10(a). The result of a test calculation is shown in Fig. 10(b). We put a singlet born RP and calculated the optimized RF $u(t)$ for initial 0.5 μs targeting T_+ and T_- spin states

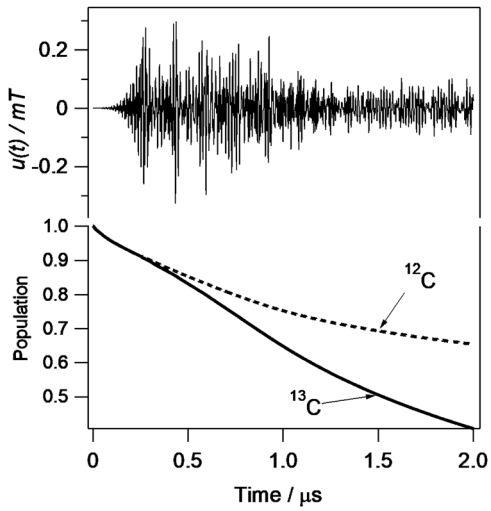


FIG. 9. Calculated isotope selective RF field $u(t)$ and the decay kinetics of RPs containing ^{12}C (broken line) and ^{13}C (solid line), respectively.

14 May 2026 17:45:38

Published under license by AIP Publishing

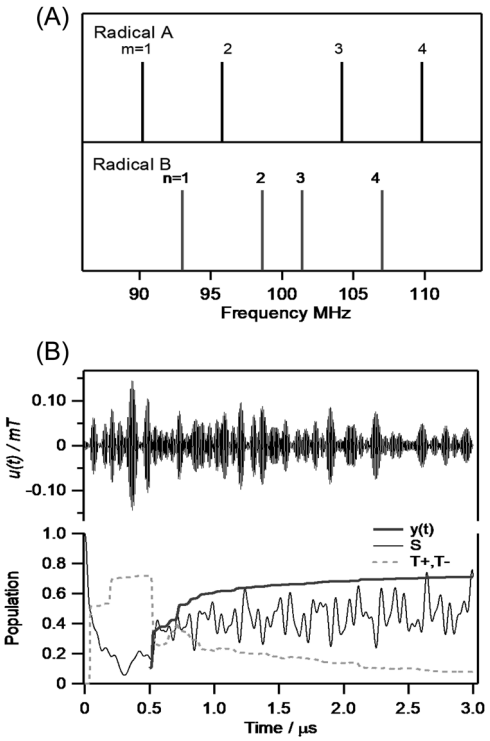


FIG. 10. Result of the coherent control calculation. Details are given in the text. (a) Stick diagram of the resonance frequencies for model RP system. Two nuclear spins couple with the electron spin in Radical A with HFCs, 0.2, 0.5 mT, and two nuclei couple with Radical B having HFCs, 0.2, 0.3 mT. (b) Optimized RF field $u(t)$ for coherent control and time evolution of the performance index $y(t)$ (thick gray line) and population of singlet state (black solid line) and T_+, T_- states (gray broken line).

$$|P_k\rangle_L = |P_{T_+}\rangle_L + |P_{T_-}\rangle_L. \tag{21}$$

It is the Liouville space representation of Eq. (10). In Fig. 10(b), the singlet character of the RP dropped to around 10% and the target population was about 70%. At $t = 0.5$ μs, we switched the target state from (21) to $|S\rangle_L$ at $t_f = 3.0$ μs. The moving target can be calculated by

$$|P_k(t)\rangle_L = \exp\left\{i\hat{\hat{L}}_0^k\left(t_f - t\right)\right\}|S\rangle_L. \tag{22}$$

Though the following dynamics were complicated, the $y(t)$ for the moving target calculated by Eqs. (8) and (22) monotonically increased as shown by the solid line in Fig. 10(b). At $t = 3.0$ μs, almost 70% of the RP becomes pure singlet state. At $t = 0.5$ μs, only about 70% of RP were in T_+ and T_- states. This imperfection of the spin polarization would be the reason for the 70% of the fidelity in the coherent control optimization.

IV. SUMMARY

In the present paper, the calculation of the optimized RF field for reaction control was demonstrated. At first, we reformulated the local optimization method of Ref. 25 to apply for the dynamics of RPs in the Liouville space. The Liouville space presentation enables us to extend the theory to the system that contains decay kinetics and/or spin relaxations. In the present paper, we have only showed the calculation including spin selective chemical reaction. The optimization including the spin relaxation must be more challenging than the conventional pure quantum mechanical optimizations. However, the calculation with spin relaxation provides more realistic RP dynamics.

The local optimization method works well even in rather complicated multi-sub-ensemble systems under the high field approximation. The high field approximation is so crude that it is not always perfectly working in the low field regime. Of course, this problem can be solved by moving to a higher frequency AWG source under a high magnetic field. Even in such a system, the calculation should not be so different from the present RF calculation because one can easily move to the rotating coordinate of the central microwave frequency. The frequency modulation of the microwave field can be calculated in a similar RF frequency range in the rotating coordinate, and the span of the modulation should be the scale of the hyperfine structures of the RPs.

It was shown that the selection of the nuclear subensembles are achieved by the useful parameters q_k. Subensemble control can be used not only for the nuclear configuration but also for the anisotropic control of RP chemical reactions. Anisotropic MFE on a RP reaction has gained great attention due to the relationship with the chemical compass[45] for animal navigation. The disruption of bird orientation by the broadband RF field is known,[9] but the optimized RF may stimulate the anisotropic response of RP reactions.

The present local optimization method may be less sophisticated than the global optimization method such as the GRAPE algorithm.[28] However, indeed, in our personal computer (PC)-level calculation, we could not apply GRAPE to the RP system having a typical number of nuclear spin configurations because of a large consumption of memory and central processing unit (CPU) time. Therefore, we could not compare our method with GRAPE directly. However, we conclude that our calculation by local optimization is practically light and easy, even for rather large dimensional quantum systems such as radical pairs coupled with multinuclear spins and useful for the manipulation of a system for optimizing the various effects in spin chemistry.

SUPPLEMENTARY MATERIAL

See the supplementary material for (1) calculation for SNP optimization for the selection of the nuclei with a larger hyperfine coupling constant (a = 0.3 mT) and (2) structure of the radical pair used for the isotope selective calculation and hyperfine coupling constants.

ACKNOWLEDGMENTS

The authors are grateful to Dr. Lewis M Antill (Saitama University) for the stimulating discussion and comments. K.M. acknowledges the financial support from JSPS KAKENHI (Grant No. 18H01184) and the "Lab-to-Lab program" in Saitama University for funding his travel to SCM 2019.

14 May 2026 17:45:38

Published under license by AIP Publishing

REFERENCES

[1] U. E. Steiner and T. Ulrich, Chem. Rev. **89**(1), 51–147 (1989).
[2] K. L. Ivanov, A. Wagenpfahl, C. Deibel, and J. Matysik, Beilstein J. Nanotechnol. **8**, 1427–1445 (2017).
[3] Y. Kobori, T. Ako, S. Oyama, T. Tachikawa, and K. Marumoto, J. Phys. Chem. C **123**(22), 13472–13481 (2019).
[4] E. A. Nasibulov, J. Behrends, L. V. Kulik, and K. L. Ivanov, Z. Phys. Chem. **231**(2), 269–291 (2017).
[5] Y. Sakaguchi, Y. Iwasaki, H. Okimi, K. Fukuno, M. Asahi, and M. Matsumura, Mol. Phys. **104**(10-11), 1719–1725 (2006).
[6] T. Ritz, S. Adem, and K. Schulten, Biophys. J. **78**(2), 707–718 (2000).
[7] T. Ritz, R. Wiltschko, P. J. Hore, C. T. Rodgers, K. Stapput, P. Thalau, C. R. Timmel, and W. Wiltschko, Biophys. J. **96**(8), 3451–3457 (2009).
[8] C. T. Rodgers and P. J. Hore, Proc. Natl. Acad. Sci. U. S. A. **106**(2), 353–360 (2009).
[9] S. Engels, N. L. Schneider, N. Lefeldt, C. M. Hein, M. Zapka, A. Michalik, D. Elbers, A. Kittel, P. J. Hore, and H. Mouritsen, Nature **509**(7500), 353–356 (2014).
[10] M. R. Wasielewski, C. H. Bock, M. K. Bowman, and J. R. Norris, Nature **303**(5917), 520–522 (1983).
[11] S. N. Batchelor, K. A. Mclauchlan, and I. A. Shkrob, Mol. Phys. **75**(3), 501–529 (1992).
[12] S. N. Batchelor, K. A. Mclauchlan, and I. A. Shkrob, Mol. Phys. **75**(3), 531–561 (1992).
[13] K. Enjo, K. Maeda, H. Murai, T. Azumi, and Y. Tanimoto, J. Phys. Chem. B **101**(50), 10661–10665 (1997).
[14] K. Maeda, Y. Araki, K. Enjo, T. Etoh, H. Murai, and T. Azumi, Appl. Magn. Reson. **12**(4), 431–439 (1997).
[15] K. Maeda, Y. Araki, Y. Kamata, K. Enjo, H. Murai, and T. Azumi, Chem. Phys. Lett. **262**(1-2), 110–114 (1996).
[16] T. Itoh, A. Matsuyama, K. Maeda, and H. Murai, Chem. Phys. Lett. **333**(3-4), 242–247 (2001).
[17] A. Matsuyama and H. Murai, J. Phys. Chem. A **106**(10), 2227–2231 (2002).
[18] A. Matsuyama, K. Maeda, and H. Murai, J. Phys. Chem. A **103**(21), 4137–4140 (1999).
[19] H. Murai, A. Matsuyama, T. Ishida, Y. Iwasaki, K. Maeda, and T. Azumi, Chem. Phys. Lett. **264**(6), 619–622 (1997).
[20] Y. Konishi and M. Okazaki, Appl. Magn. Reson. **14**(1), 131–141 (1998).
[21] M. Okazaki and K. Toriyama, J. Phys. Chem. **99**(47), 17244–17250 (1995).
[22] M. Okazaki and T. Shiga, Nature **323**(6085), 240–243 (1986).
[23] Y. Sakaguchi, Chem. Phys. Lett. **345**(1-2), 51–56 (2001).
[24] K. Maeda, C. J. Wedge, J. G. Storey, K. B. Henbest, P. A. Liddell, G. Kodis, D. Gust, P. J. Hore, and C. R. Timmel, Chem. Commun. **47**(23), 6563–6565 (2011).
[25] M. Sugawara, J. Chem. Phys. **118**(15), 6784–6800 (2003).
[26] R. Kosloff, S. A. Rice, P. Gaspard, S. Tersigni, and D. J. Tannor, Chem. Phys. **139**(1), 201–220 (1989).
[27] D. J. Tannor and S. A. Rice, J. Chem. Phys. **83**(10), 5013–5018 (1985).
[28] N. Khaneja, T. Reiss, C. Kehlet, T. Schulte-Herbruggen, and S. J. Glaser, J. Magn. Reson. **172**(2), 296–305 (2005).
[29] C. D. Buckley, D. A. Hunter, P. J. Hore, and K. A. Mclauchlan, Chem. Phys. Lett. **135**(3), 307–312 (1987).
[30] K. Maeda, J. G. Storey, P. A. Liddell, D. Gust, P. J. Hore, C. J. Wedge, and C. R. Timmel, Phys. Chem. Chem. Phys. **17**(5), 3550–3559 (2015).
[31] R. Haberkorn, Mol. Phys. **32**(5), 1491–1493 (1976).
[32] K. L. Ivanov, M. V. Petrova, N. N. Lukzen, and K. Maeda, J. Phys. Chem. A **114**(35), 9447–9455 (2010).
[33] J. A. Jones, K. Maeda, U. E. Steiner, and P. J. Hore, Chem. Phys. Lett. **508**(1-3), 184–185 (2011).
[34] J. A. Jones, K. Maeda, and P. J. Hore, Chem. Phys. Lett. **507**(4-6), 269–273 (2011).
[35] V. R. Gorelik, K. Maeda, H. Yashiro, and H. Murai, J. Phys. Chem. A **105**(34), 8011–8017 (2001).
[36] V. F. Tarasov, E. G. Bagryanskaya, I. A. Shkrob, N. I. Avdievich, N. D. Ghatlia, N. N. Lukzen, N. J. Turro, and R. Z. Sagdeev, J. Am. Chem. Soc. **117**(1), 110–118 (1995).
[37] R. Z. Sagdeev and E. G. Bagryanskaya, Pure Appl. Chem. **62**(8), 1547–1556 (1990).
[38] N. I. Avdievich, E. G. Bagryanskaya, Y. A. Grishin, and R. Z. Sagdeev, Chem. Phys. Lett. **155**(2), 141–145 (1989).
[39] E. G. Bagryanskaya, Y. A. Grishin, R. Z. Sagdeev, and Y. N. Molin, Chem. Phys. Lett. **128**(4), 417–419 (1986).
[40] N. J. Turro and B. Kraeutler, J. Am. Chem. Soc. **100**(23), 7432–7434 (1978).
[41] E. N. Step, A. L. Buchachenko, and N. J. Turro, Chem. Phys. **162**(1), 189–204 (1992).
[42] M. Wakasa, H. Hayashi, T. Kobayashi, and T. Takada, J. Phys. Chem. **97**(51), 13444–13446 (1993).
[43] M. Wakasa, H. Hayashi, K. Ohara, and T. Takada, J. Am. Chem. Soc. **120**(13), 3227–3230 (1998).
[44] T. Mizuno, M.Sc. thesis, Saitama University, 2019; T. Yago, personal communication (2019).
[45] K. Maeda, K. B. Henbest, F. Cintolesi, I. Kuprov, C. T. Rodgers, P. A. Liddell, D. Gust, C. R. Timmel, and P. J. Hore, Nature **453**(7193), 387–390 (2008).

14 May 2026 17:45:38

Published under license by AIP Publishing